

3. TESTS FOR FUNCTIONAL GROUPS

The tests for functional groups given below will help you to determine the presence or absence of a particular functional group. An understanding of these tests is useful to identify the class to which an unknown organic compound belongs. Procedures for carrying out the tests for functional groups have been described under the different classes of organic compounds. Record all observations for each reaction in your record book immediately.

Preliminary classification

Physical characteristics of the unknown compound should be noted carefully. These include its physical states, (Liquid, solid, crystalline form etc.) its color and odor. Many compounds have characteristic colors or odors. Compounds with a high degree of conjugation are frequently yellow or red. Amines often have a fishlike odor. [Caution! Many compounds have distinctly unpleasant or nauseating odors. Some have corrosive vapors. Therefore any unknown substances should be sniffed with the greatest caution]

Tests for unsaturation

1. Br₂/ CCl₄ test

Procedure: dissolve few crystals (50 mg) of a solid unknown or 2 drops of a liquid unknown in CCl₄ or CH₂Cl₂ (1 ml) and add a solution of Br₂ in CCl₄/ CH₂Cl₂ drop wise with shaking, until bromine color persists.

The test is positive if more than 5 drops of the bromine solution are needed so that the color remains for 1 minute. Usually, many drops of the bromine solution will be needed if unsaturation is present.

Aromatic compounds either do not react with Br₂/CCl₄ reagent or they react by substitution. Only aromatic rings with activating groups as substituents (-OH, -OR or NR₂) give this substitution reaction. When substitution occurs, HBr gas is also evolved and can be detected by a moistened litmus or pH paper. (See bromine water test for phenols)

2. Potassium permanganate (Baeyer test)

A positive test is given by compounds with double and triple bonds but aromatic rings do not react. The test depends on the conversion of the purple MnO₄⁻ to a brown precipitate MnO₂ following the oxidation of an unsaturated compound. Other easily oxidized compounds such as aldehydes, some alcohols, phenols and aromatic amines also give a positive test with KMnO₄.

Procedure: dissolve the unknown compound in water or in 95 % ethanol and slowly add a 1% aq. solution of KMnO₄ drop by drop with shaking. In a positive test, the purple color of the reagent is discharged, and a brown precipitate of manganese dioxide is formed.

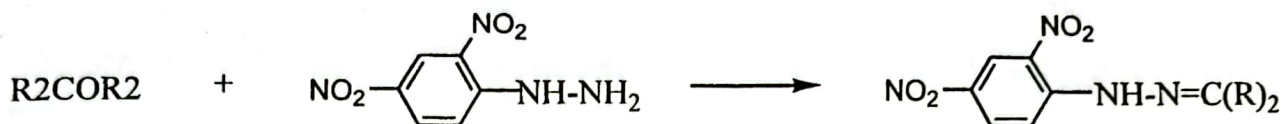
Tests for aldehydes and ketones

Aldehydes and ketones contain a carbonyl group and often react in a similar way. However, aldehydes usually react faster than ketones with the same reagent because of lesser crowding around the carbonyl carbon atom in aldehydes. Aldehydes are also more

readily oxidized than ketones. Note any odor, solubility in water and action on litmus paper.

1. Brady's reagent

Aldehydes and ketones condense with hydrazine ($\text{NH}_2\text{-NH}_2$) and substituted hydrazines to form hydrazones and substituted hydrazones. Even lower aliphatic aldehydes and ketones give orange- to- red precipitates with 2,4-dinitrophenylhydrazine (Brady's reagent)

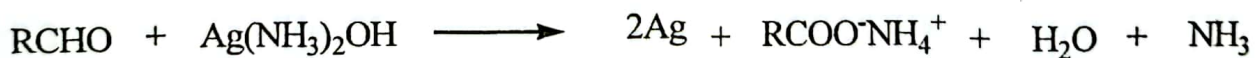


Procedure: Add one drop of the liquid unknown to Brady's reagent (1mL). If the unknown is solid, dissolve a few crystals (10 mg) in a minimum amount of 95% ethanol before adding it to the reagent. Shake the mixture vigorously. Most carbonyl compounds give yellow to red precipitates immediately. However some compounds will require up to 15 minutes, or even gentle heating, to give a precipitate.

2. Tollen's reagent/ Ammoniacal silver nitrate

[This test should be used only if it has already shown that the unknown compound is either an aldehyde or ketones.] The reagent must be prepared immediately before use and all residues disposed off immediately after use. On standing, reagents tends to form silver fulminate, a very explosive substance.

Procedure: Add aq. NaOH (1 drop) to AgNO_3 solution (about 2 mL) contained in a test tube. Add NH_4OH drop wise until the Ag_2O formed is just dissolved. Add 2-3 drops of the liquid aldehyde or few crystals of the solid aldehyde dissolved in alcohol to this solution. Allow to stand at room temperature. If a silver mirror is not formed, warm gently on a warm water bath. Aldehydes, reducing sugars and other reducing agents reduce Ag^+ to metallic silver, which forms a mirror on the inner wall of the test tube.



3. Fehling's test

Fehling's solution is made when required by mixing the two Fehling's solutions, Fehling's A and Fehling's B, in equal proportions.

Procedure: Mix equal volumes of Fehling's A and Fehling's B solutions and add a few drops or crystals of the unknown carbonyl compounds. Boil the mixture gently. Note the taken to form reddish brown precipitate of CuO_2 . Aliphatic aldehydes reduce Fehling's solution rapidly and the reduction is slow with aromatic aldehydes. Reducing sugars also rapidly reduce Fehling's solution.

4. Iodoform test

An alkaline solution of sodium hypoiodite (from NaOH and I₂) converts acetaldehyde, methyl ketones and aliphatic alcohols of the general formula, RCH(OH)CH₃ in to iodoform (CHI₃). Iodoform is a yellow solid with a characteristic smell.

Procedure: Place aqueous NaOH solution (6-8 drops) and an equal volume of I₂/KI solution in a test tube. Add the test compound (3-4 drops) and warm gently in a warm water bath. Shake test tube vigorously, cool, dilute with little water and allow to stand for 15 minutes. A yellow precipitate of iodoform with the characteristic antiseptic smell will be founded.

5. Chromic acid test (Bordwell-wellman reagent)

Procedure: Dissolve the liquid unknown (one drop) or the solid unknown (10mg) in acetone (1mL). Add several drops of the chromic acid reagent, a drop at a time with shaking. A positive test is indicated by a green precipitate or a green color of the reagent.



Primary and secondary alcohols are also oxidized by this reagent. Therefore this test is not useful in identifying aldehydes unless a positive identification of the carbonyl group has already been made.